

YUEZHEN (LILY) DONG

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RELEVANT SKILLS

Languages	Python, Java, C/C++, TypeScript/JavaScript, SQL, HTML/CSS, Ruby, Bash
Frameworks	Spring Boot, React/Next.js, Node/Express, Three.js, Flask, Pygame, Arduino, GraphQL
Developer Tools	AWS, Docker, Kubernetes, Git, Maven, Linux, PostgreSQL, AutoCAD, VSCode, IntelliJ

EDUCATION

University of Waterloo - Faculty of Engineering	Expected Graduation MAY 2028
BASc in Electrical Engineering	Waterloo, ON, CAN

ACHIEVEMENTS

IBDP Graduate CaYPT (Canadian Young Physicists' Tournament) National Champion IYPT (International Young Physicists' Tournament) Bronze Medalist FIRST Robotics Provincial Semi-Finalist

WORK EXPERIENCE

Software Engineering Intern SEP 2025 – DEC 2025
React, TypeScript, Ruby, GraphQL *Shopify*

- Built retail onboarding flows (React+TypeScript; Rails/GraphQL) from install to first in-person sale; cut time-to-first-sale by **35%** and boosted day-0 adoption for **500+** merchants.
- Developed cross-surface flow across POS & Admin (Rails/GraphQL, HTML/JS, React) with staged rollout to 100% stores; achieved **98%** on-time acceptance (↑22pp), **-60%** tickets, and **95%** test coverage with runbooks.

Web Developer Coop JAN 2025 – APR 2025
Java, Spring Boot, Maven, Apache POI, Git *AGF Investments*

- **(In production — company-wide)** Independently developed a **Java-based** ScoreCard application with file archiving, large XLSX handling via **Apache POI streaming**, data validation/transformation, and a properties-based configuration system; now used in **daily reporting workflows across 4 departments**, saving **20+ staff hours/week**.
- **(In production — company-wide)** Delivered an end-to-end **Spring Boot** reporting pipeline to ingest XLSX, filter by date/time, archive inputs, generate Excel/CSV outputs, and automated email distribution under **Maven/Git**.
- Built **Pega** scenario tests and **Selenium** UI/service tests covering 90% of core business flows, reducing regression defects by **40%** and enabling releases to ship **2x faster**.

Website Developer DEC 2020 – DEC 2023
Java, HTML/CSS, JavaScript *Beyond the Wards / Kindness Catalogue*

- Designed and deployed a dynamic website using **Java** backend and **HTML/CSS, JavaScript** frontend, ensuring mobile responsiveness and smooth interaction; maintained uptime of **99.9%** and supported **5,000+ monthly visitors**.

PERSONAL PROJECTS

Ecoland: AI Ecosystem Simulator
Java, JavaFX, OOP, AI, GeneticAlgorithms

- Built a **Java** simulator with custom **AI algorithms**, pathfinding, and genetic algorithms to model species interactions.
- Employed **concurrency**, **data structures**, and **OOP** principles for large-scale simulations, enabling real-time monitoring, performance tuning, and modular feature expansions.

3D Data Sandbox
Electron, Node.js, Three.js, PapaParse, JavaScript

- Built a cross-platform **Electron** desktop application using **Three.js** and **Node.js/PapaParse** for offline 3D visualization, parsing, and analysis of local CSV/JSON datasets.
- Implemented scatter, grid, and K-means layouts with outlier detection, correlation lines, and user-friendly UI controls for filtering, highlighting, and exporting.

Puppy Portal Puzzle: Time-Trotting Dogventure
Python, Pygame, Pathfinding, Time-Manipulation, Puzzle Architecture

- Developed a controlled **time-travel** mechanic with simultaneous clones, integrating **BFS pathfinding** for AI enemies.
- Structured the game using **Python OOP** and event-driven architecture, implementing multi-layer switch logic, portal systems, and crate interactions while ensuring seamless performance under increasing complexity.